Corpus
no queries,
no labels

1 Write Queries

An LLM writes a query from documents sampled from the corpus

task instruction (optional) → **LLM** → *What non-genetic factors can trigger cancer development in the presence of oncogenic mutations?*

1 of 100 per task

2 Retrieve

Every model returns its top 10 for each query

	Docs			
Qwen3-8B	✓	✓	✓	✗
Linq-Embed-Mistral	✓	✓	✓	✗
all-MiniLM-L6-v2	✓	✗	✗	✓
baseline-bm25s	✓	✓	✓	✗

✓ = in the model's top 10 (4 of 28 models shown)

3 Judge the results, in one of two ways

Pointwise grade each document once

Pool

Every document any model returned

$D_j = \bigcup_m \text{top10}_m(q_j)$

87 documents here

Grade

The judge sees the query and one document

q_j + → **Judge** → 0 1 2 3 4 5

$g_{ij} = \sum_k k \cdot p(k \mid q_j, d_i) = 4.3$

Score

nDCG@10 with the grades as labels

$$\text{nDCG@10} = \frac{\sum_{r \le 10} g_r \,/\, \log_2(r+1)}{\text{the same for the 10 best in } D_j}$$

A 0.81 B 0.35

Judge calls per query: one per pooled document (87 here)

Pairwise compare two models' lists

Pair Up

Every pair of models

378 pairs of 28 models,
or a random subset

Compare

The judge sees the query and both top 10s

q_j → **Judge** →
A first: $P(\text{A better}) = 0.97$
B first: $P(\text{A better}) = 0.62$

s_{AB} = mean of both orders = 0.80

each document cut to 4,000 characters

Score

Bradley–Terry fit to all pair scores

$$P(m \text{ beats } n) = \frac{\pi_m}{\pi_m + \pi_n}$$

$\text{Elo}_m = 400 \log_{10} \pi_m$, mean 1000

Judge calls per query: two per pair (756 for all pairs)

4 Rank

1 Qwen3-8B
2 Qwen3-4B
3 Linq-Embed-Mistral
⋮
25 all-MiniLM-L6-v2

pointwise:
$\rho = 0.95$ with MTEB